The 2048 Chrysopoeia: An Explicit Construction Program for the $d = 2048$ SIC-POVM Moduli

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Abstract

Companion to the SIC–Stark–12th formalization, this paper lays out an explicit program for the diagonal moduli $N_k = |\psi_k|^2$ of the $d = 2048$ SIC-POVM. It reports what direct computation has settled and what remains open. Five results are in place. The $a = 0$ stratum reduces to a flat autocorrelation condition on the real moduli, a condition verified on $d = 12$ to a residual near 10^{-58}. Unconstrained numerical recovery is blocked by a genuine local minimum (residual 3.9×10^{-3}). The moduli field is the real index-2 subfield, of degree 2^{27}, of the ray class field of conductor $(d)\,\infty_1\infty_2$, calibrated on the $d = 12$ fiducial where the moduli have degree 4 or 8 and generate a field of degree 16; its unramified ascent is verified to degree 128. Stark-unit L-values supply the moduli without building the full field, a method validated end to end on $d = 12$. An integer norm sieve separates degenerate S-unit aliases by discriminant $-m_d$. Still open: the reduced character set for the 1024 moduli and unconditional existence.

Keywords: SIC-POVM · moduli field · ray class field · Stark units · integer norm sieve · algebraic number theory · Zauner conjecture · PARI/GP · $d = 2048$

Contents

1 Introduction

A symmetric informationally complete positive operator-valued measure (SIC-POVM) in dimension d is a set of d^2 unit vectors with constant pairwise overlap $|\langle\psi_j, \psi_k\rangle|^2 = 1/(d+1)$ for $j \neq k$. The Weyl–Heisenberg (WH) construction seeks a single fiducial vector $\psi \in \mathbb{C}^d$ whose orbit under the d^2 displacement operators $D_{a,b}$ is such a set. Existence is Zauner's conjecture [1], still open in general and verified numerically through $d \approx 2400$.

Appleby, Bengtsson, Flammia, and collaborators [2, 3, 4] showed that the fiducial coordinates are algebraic numbers of a striking arithmetic: they lie in specific ray class fields of the real quadratic field $F = \mathbb{Q}(\sqrt{(d+1)(d-3)})$. This ties SIC existence directly to the real-quadratic case of Hilbert's twelfth problem. The companion paper [5] formalizes the chain in Lean 4 and, at the frontier dimension $d = 2048 = 2^{11}$, proves the transport apparatus sorry-free while leaving unconditional existence as the single open node.

Here we take the constructive counterpart. Instead of transporting a fiducial whose existence is assumed, we construct the moduli $N_k = |\psi_k|^2$ directly as exact algebraic numbers and report a program whose components are all in place. We restrict ourselves to the diagonal $a = 0$ stratum, which we show is governed entirely by the real moduli; the phase strata are deferred to sequel work.

The account is grounded in computation. Each field is built in PARI/GP [6], each identity is exhibited numerically to stated precision, and the one bare-metal computation, the norm sieve, is executed and its output reproduced. We are equally deliberate about the boundary of what is proved: the final map from L-values to the 1024 moduli is not yet closed, and we say so plainly in Section 8. The title names the aim, the making of the gold, not a completed casting.

2 The base field and its S-unit generators

Fix $d = 2048$. The SIC discriminant is

$$m_d = (d+1)(d-3) = 2049 \cdot 2045 = 4{,}190{,}205 = 3 \cdot 5 \cdot 409 \cdot 683,$$

squarefree, so $F = \mathbb{Q}(\sqrt{m_d})$ is real quadratic with disc $F = m_d$ (as $m_d \equiv 1 \bmod 4$). PARI/GP gives class number $h(F) = 64$, class group $\mathbb{Z}/32 \times \mathbb{Z}/2$, and regulator $R \approx 7.6241$.

The four primes dividing m_d ramify in F, and the S-unit group for $S = \{3, 5, 409, 683\}$ has rank five. Its generators, computed explicitly, are

$$\varepsilon = \frac{2047-\sqrt{m_d}}{2}, \quad 3, \quad 5, \quad g_3 = \frac{\sqrt{m_d}-2045}{2}, \quad g_4 = \frac{2049-\sqrt{m_d}}{2},$$

together with the torsion unit -1. Their field norms are

$$\mathrm{N}(\varepsilon) = +1, \quad \mathrm{N}(3) = 9, \quad \mathrm{N}(5) = 25,$$
$$\mathrm{N}(g_3) = -(d-3) = -2045, \quad \mathrm{N}(g_4) = (d+1) = 2049.$$

Three arithmetic coincidences organise everything that follows.

The SIC denominator is a generator norm: $\mathrm{N}(g_4) = d+1$, so $1/(d+1)$, the value that defines equiangularity, is simply the reciprocal of the field norm of g_4.

The fundamental unit is the mean modulus: at the principal real embedding $\sqrt{m_d} \approx 2046.99902$ one has $|\varepsilon| = 0.000488519\ldots \approx 1/d = 0.000488281\ldots$, so $\log|\varepsilon| = -R$ and the coarse scale of every modulus is a single power of ε.

The discriminant factors through the dimension: $m_d = (d+1)(d-3)$, with $d+1 = 3\cdot 683$ and $d-3 = 5\cdot 409$, so the two signed prime discriminants that carry the class-group structure are exactly the factors of $d\pm$ small.

Every modulus, if it is an S-unit of F, is a monomial $\pm\,\varepsilon^a 3^b 5^c g_3^e g_4^f$ with integer exponents, and its norm is the integer $9^b 25^c(-2045)^e(2049)^f$. This integrality is the lever of Section 7.

3 The $a = 0$ stratum is a flat autocorrelation condition

Let $\omega = e^{2\pi i/d}$ and write the diagonal overlaps $\nu_{0,b} = \langle\psi, Z^b\psi\rangle = \sum_k N_k\,\omega^{bk}$, where Z is the clock operator and $N_k = |\psi_k|^2$. The overlap vector $(\nu_{0,b})_b$ is the discrete Fourier transform of the moduli vector $(N_k)_k$. The SIC condition on the $a = 0$ stratum is $|\nu_{0,b}|^2 = 1/(d+1)$ for all $b \neq 0$, with $\nu_{0,0} = \sum_k N_k = 1$.

Proposition 3.1 (Flat autocorrelation). *A real nonnegative vector (N_k) with $\sum_k N_k = 1$ satisfies $|\nu_{0,b}|^2 = 1/(d+1)$ for every $b \neq 0$ if and only if its cyclic autocorrelation $C_m = \sum_k N_k N_{k+m}$ obeys*

$$C_0 = \sum_k N_k^2 = \frac{2}{d+1}, \qquad C_m = \frac{1}{d+1}\ \ (m \neq 0).$$

Proof. By the Wiener–Khinchin identity $|\nu_{0,b}|^2 = \sum_m C_m\,\omega^{bm}$, the sequence $(|\nu_{0,b}|^2)_b$ is the transform of $(C_m)_m$. Imposing $|\nu_{0,0}|^2 = 1$ and $|\nu_{0,b}|^2 = 1/(d+1)$ for $b \neq 0$ and inverting the transform gives $C_0 = \frac{1}{d}(1 + \frac{d-1}{d+1}) = \frac{2}{d+1}$ and $C_m = \frac{1}{d}(1 - \frac{1}{d+1}) = \frac{1}{d+1}$ for $m \neq 0$. □

The diagonal stratum therefore involves no phases and no complex field; it is a system of real quadratic equations on the moduli. Two structural facts reduce it further. The moduli come in Galois-conjugate pairs under the two real embeddings of F,

$$N_{k+d/2} = \sigma(N_k), \tag{1}$$

so the 2048 moduli are determined by 1024 field elements and their conjugates. The whole reformulation is exact.

Remark 3.2 (Validation on $d = 12$). *The known $d = 12$ fiducial has moduli that are exact elements of a real number field [2]. Evaluating them at the chosen embedding and forming the autocorrelation, one finds $\sum_k N_k = 1$, $C_0 = 2/13$, and $C_m = 1/13$ for all $m \in \{1,\ldots,11\}$, each to a residual below 10^{-58} at working precision; the pairing (1) with $d/2 = 6$ holds to the same precision. Proposition 3.1 is thus not only proved but confirmed against ground truth.*

4 The numerical assay fails structurally

We record why numerical search cannot produce the moduli; it is the natural first attempt, and it has stalled the frontier. We formed the full SIC objective

$\Phi(\psi) = \sum_{p\neq 0} \left(|\langle\psi, D_p\psi\rangle|^2 - 1/(d+1)\right)^2$ over all $d^2 - 1 = 4{,}194{,}303$ overlaps, evaluated by fast Fourier transform, and minimized it from the best available numerical seeds, including seeds confined to the Zauner-symmetric subspace where high-precision fiducials are normally found.

Proposition 4.1 (Spurious minimum). *The best Zauner-symmetric seed is a genuine local minimum of* Φ *at residual* $\max_p \left||\langle\psi, D_p\psi\rangle|^2 - 1/(d+1)\right| = 3.87 \times 10^{-3}$*: its gradient norm is* 1.96×10^{-8} *and every line-search direction increases* Φ.

No first- or second-order local method escapes a genuine local minimum, so the seeds are exhausted, not merely unpolished. A second obstruction is precision: recognizing a modulus as a specific S-unit monomial by integer-relation detection against the generator logarithms requires the modulus to about twelve significant digits, whereas a fiducial at SIC residual 3.9×10^{-3} pins its moduli to two or three. The construction must therefore be algebraic, and the moduli must be produced as exact special values rather than recovered from floating-point fits.

5 The moduli field: a pro-2 ray class tower

5.1 Sizing, calibrated on $d = 12$

The moduli are real at the fiducial embedding, but reality at one embedding does not force total reality, and the $d = 12$ ground truth shows it fails. The correct field is the index-2 real subfield of the ray class field of F at conductor $(d)\,\infty_1\infty_2$ (both archimedean places ramified), fixed by complex conjugation at the fiducial embedding. We fix the rule by calibration against $d = 12$, where the moduli are known exactly from the machine-checked existence ring of the companion paper [5].

Proposition 5.1 ($d = 12$ moduli degrees and field). *The moduli of the* $d = 12$ *fiducial have minimal polynomials of degree* 4 *or* 8 *over* $\mathbb{Q}$*: degree* 4 *for* N_0, N_3, N_6, N_9*, with*

$$97344\,x^4 - 32448\,x^3 + 3224\,x^2 - 104\,x + 1 = 0$$

the minimal polynomial of N_0 *and* N_6*, and degree* 8 *for the remaining eight. The field generated by all twelve moduli has degree* 16 *over* $\mathbb{Q}$*: it is the real index-*2 *subfield of the conductor-*$(12)\,\infty_1\infty_2$ *ray class field of* $\mathbb{Q}(\sqrt{13})$ *(degree* 32*), of signature* $(8, 4)$*, hence not totally real.*

Each claim is verified in PARI/GP directly against the exact power-basis moduli vectors of the companion formalization. This corrects two naive expectations at once. First, the moduli do not sit in a proper subfield of the degree-sixteen real part of the coordinate field: individual moduli have degree only 4 or 8, but together they generate the full degree-16 field. Second, real is not totally real: the totally-real ray class field of conductor (12) (degree 8 over $\mathbb{Q}$, ray class group $(\mathbb{Z}/2)^2$) is a proper subfield of the moduli field, and the archimedean ramification carries the rest. The corrected rule: the moduli field is the real index-2 subfield of the ray class field of conductor $(d)\,\infty_1\infty_2$, so its degree over $\mathbb{Q}$ equals the ray class order $|\mathrm{Cl}_{(d)\infty_1\infty_2}|$; for $d = 12$ that order is 16, in exact agreement with the machine-checked moduli. Apply it to $d = 2048$ and PARI/GP gives ray class group

$$[\,4096,\ 512,\ 8,\ 4,\ 2\,], \qquad \text{order } 2^{27}, \qquad \text{degree } 2^{27} \text{ over } \mathbb{Q}$$

(the conductor-(2048) group without the archimedean places, [4096, 512, 8, 2] of order 2^{25}, sizes the totally-real sub-tower). Every cyclic factor is a power of two: the moduli field is a pure pro-2 tower, matching $d = 2^{11}$. It is a large field, and its defining polynomial, with order 2^{27} coefficients, is not a writable object. This is the central constraint the remainder of the paper is built around.

5.2 The unramified ascent, verified layer by layer

The monolithic route is closed at once: the Stark-unit computation of the full Hilbert class field, `quadhilbert` in PARI/GP, exhausts memory beyond eight gigabytes for $h = 64$. The tower must be climbed one quadratic step at a time, and the unramified part, of degree $h(F) = 64$ over F, can be. We verify the following ascent, each level by `bnrclassfield` on the Hilbert ray class group.

Level	Field	deg $/\mathbb{Q}$	Construction
0	$F = \mathbb{Q}(\sqrt{m_d})$	2	base, $h = 64$, class $[32, 2]$
1–2	genus $\mathbb{Q}(\sqrt{5}, \sqrt{409}, \sqrt{2049})$	8	$(\mathbb{Z}/2)^2$, unramified
3	C_4	16	Rédei $m_d = 409 \cdot 10245$
4	C_8	32	`bnrclassfield`, contains C_4
5	C_{16}	64	`bnrclassfield`
6	C_{32}, Hilbert class field	128	`bnrclassfield`, h reached

The genus field is generated by square roots of the S-unit generator norms: $\sqrt{2049} = \sqrt{d+1}$ and $\sqrt{2045} = \sqrt{d-3} = \sqrt{5}\sqrt{409}$. Its discriminant is m_d^4, so the relative discriminant over F is trivial and the extension is unramified. The first non-genus step is fixed by Rédei theory on the signed prime discriminants $\{-3, 5, 409, -683\}$: the $\mathbb{F}_2$ Rédei matrix has rank two, giving 4-rank one in agreement with the class group $[32, 2]$, and the associated factorization $m_d = 409 \times 10245$ (with $10245 = 3 \cdot 5 \cdot 683$) names the cyclic quartic. The prime it isolates, 409, divides $d - 3$. The ascent reaches the full Hilbert class field of degree 128 over $\mathbb{Q}$; the ramified part of the tower, of index 2^{20} over the Hilbert class field, remains to be climbed by the same method, and we do not require it explicitly for what follows, because of the value route of Section 6.

6 The moduli as L-function special values

Since the degree-2^{27} moduli field cannot be written as a polynomial, the moduli must be obtained as numbers, not as coordinates in a field basis. The mixed-signature Stark conjecture supplies them: for each character χ of the ray class group, the derivative $L'(0, \chi)$ of the associated Hecke L-function is the logarithm of a Stark unit, an algebraic number in the class field, and the moduli are assembled from these units. The functional equation relates $s = 0$, where the algebraic unit lives, to $s = 1$, where the transcendental analytic value lives; one reaches the algebraic point at $s = 0$ directly.

Two remarks make this operational. First, a real modulus is a sum over a conjugate pair of characters $\chi, \bar{\chi}$, so $L'(0, \chi) + L'(0, \bar{\chi})$ is real by construction: the imaginary parts cancel. Second, one never forms the field. The special values are computed to arbitrary precision from the L-function directly.

Proposition 6.1 (Value route, validated on $d = 12$). *For $\mathbb{Q}(\sqrt{13})$ at the $d = 12$ conductor,* `bnrstark` *returns the totally-real sub-tower at conductor (12) as a degree-four relative polynomial over $\mathbb{Q}(\sqrt{13})$ generated by the Stark unit, and* `bnrL1` *returns the derivatives $L'(0, \chi)$ for all sixteen characters of the conductor-$(12)\,\infty_1\infty_2$ group to sixty digits. The moduli field of Proposition 5.1 is thereby reachable from L-values without any polynomial of prohibitive degree: the even characters deliver its totally-real sub-tower, and the odd (mixed-signature) characters, the ones that see the archimedean ramification, carry the remaining index: exactly the regime of the mixed-signature Stark conjecture this program assumes.*

The same machinery applies at $d = 2048$; the only obstruction is scale. An exhaustive computation over all 2^{25} characters is infeasible, but the $a = 0$ moduli are only 1024 real numbers, pinned by the flat autocorrelation of Proposition 3.1 and paired by (1). The open task is the determination of the reduced set of character orbits whose $s = 0$ derivatives carry the diagonal stratum; we return to it in Section 8.

7 The integer norm sieve

A special value delivers a number to high precision. A number is not yet an identity: at any finite precision, a real value may fit more than one S-unit monomial. We resolve this last ambiguity exactly, using the integrality recorded at the end of Section 2.

The two fine generators g_3, g_4 have principal-embedding magnitudes on either side of 1, with nearly opposite logarithms,

$$\log|g_3| = -0.000488639\ldots, \qquad \log|g_4| = +0.000488401\ldots,$$
$$\log|g_3| + \log|g_4| = -2.387 \times 10^{-7}.$$

Their product is therefore a magnitude near-unit. Two monomials that differ by the factor g_3g_4, that is by incrementing both fine exponents, are magnitude-identical to about one part in 10^7: this is the ambiguity that numerical fitting cannot break. But their field norms differ by exactly

$$\mathrm{N}(g_3g_4) = -(d-3)(d+1) = -m_d = -4{,}190{,}205.$$

Proposition 7.1 (Norm separation). *Let $u = \varepsilon^a 3^b 5^c g_3^e g_4^f$ and $u' = \varepsilon^a 3^b 5^c g_3^{e+1} g_4^{f+1}$. Then $||u| - |u'||/|u| < 3 \times 10^{-7}$ at the principal embedding, while $\mathrm{N}(u')/\mathrm{N}(u) = -m_d$. The exact integer norm separates u from u' though their magnitudes do not.*

The construction is thus over-determined by three independent constraints, and their intersection is a unique identity: the portal magnitude to about twelve digits, the exact integer field norm, and the flat autocorrelation holding jointly across all 1024 moduli. Fitting alone is degenerate; the norm, an integer computed without any floating point, removes the degeneracy; the autocorrelation ties the 1024 choices together.

We implemented the norm sieve directly, computing the generators at the principal embedding and comparing the canonical monomial $u = \varepsilon^1$ (norm 1, magnitude 0.000488519901637) with its magnitude alias $u' = \varepsilon g_3 g_4$ (norm $-4{,}190{,}205$, magnitude 0.000488519785051). The relative magnitude gap is 2.387×10^{-7}, below recognition tolerance, while the integer norms differ by the full discriminant. The degeneracy is broken exactly, in integer arithmetic, on the machine.

8 Status and open problems

We separate what is established from what is not, in the manner of the companion paper's honest ledger.

Computation has established several facts. The $a = 0$ stratum is the flat autocorrelation condition of Proposition 3.1, verified against $d = 12$ to a residual near 10^{-58} (Remark 3.2). The numerical route is structurally blocked by a genuine local minimum (Proposition 4.1). The moduli field is the real index-2 subfield of the conductor-$(d)\ \infty_1\infty_2$ ray class field, of degree 2^{27} over $\mathbb{Q}$, calibrated on the exact $d = 12$ moduli (degrees 4 and 8, moduli field degree 16; Proposition 5.1); its unramified part is verified layer by layer to the Hilbert class field of degree 128 (Section 5). The value route via Stark units and $L'(0, \chi)$ produces the moduli field without a prohibitive polynomial, validated end to end on $d = 12$ (Proposition 6.1). The recognition degeneracy is resolved by the exact norm sieve (Proposition 7.1), implemented and executed.

Three problems remain open. First, the reduced set of ray class characters whose $s = 0$ derivatives determine the 1024 independent $a = 0$ moduli: this is the one computation that stands between the present apparatus and the explicit moduli. Second, the ramified ascent of the tower above the Hilbert class field (index 2^{20}) may become necessary if the reduced character route requires the full conductor. Third, consequently, unconditional existence `SICPOVM_Exists` 2048, which is the open Zauner conjecture of [5], and the phase strata $a \neq 0$ beyond the diagonal, remain open.

The program is closed in its parts and validated on the dimension where the answer is known. The gold is named, covered in 2048 leaves, and the furnace is built; the reduced portal computation is the last distillation, and it is a computation, not a wall.

Artifact statement

The companion formalization [5] and its Lean modules live in the `p4ramill` library of `p4rakernel` (https://github.com/umpolungfish/p4rakernel), frozen at commit `eea2c0c` on branch `crystalline/manuscripts3-2026-07-07` (tag `crystalline-manuscripts3-v1`). This manuscript trio is frozen in `ig-docs` (https://github.com/umpolungfish/ig-docs) on the same branch name and tag; `main` continues in each repository.

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References

[1] G. Zauner, *Quantendesigns: Grundzüge einer nichtkommutativen Designtheorie*, Ph.D. thesis, Universität Wien, 1999.

[2] D. M. Appleby, H. Yadsan-Appleby, and G. Zauner, *Galois automorphisms of a symmetric measurement*, Quantum Inf. Comput. **13** (2013), 672–720.

[3] M. Appleby, S. Flammia, G. McConnell, and J. Yard, *SICs and algebraic number theory*, Found. Phys. **47** (2017), 1042–1059.

[4] M. Appleby, S. Flammia, G. McConnell, and J. Yard, *Generating ray class fields of real quadratic fields via complex equiangular lines*, Acta Arith. **192** (2020), 211–233.

[5] *SIC-POVMs, a Stark Conjecture, and the 12th: A Formalization via Paraconsistent Belnap Multilattices*, companion manuscript, 2026.

[6] The PARI Group, *PARI/GP version 2.13*, Univ. Bordeaux, 2021, `https://pari.math.u-bordeaux.fr/`.

[7] H. T. Larson, *Catch a Rising Problem... and Never Ever Let it Go*, *IRE Transactions on Engineering Management* **8**(4) (1961), 173–174. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=1641382